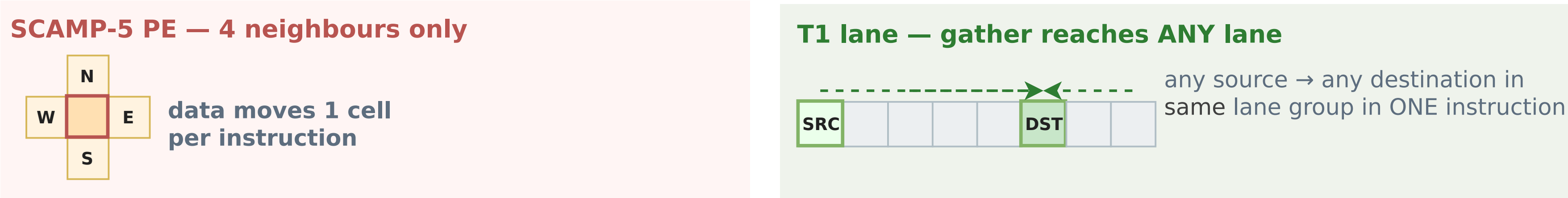
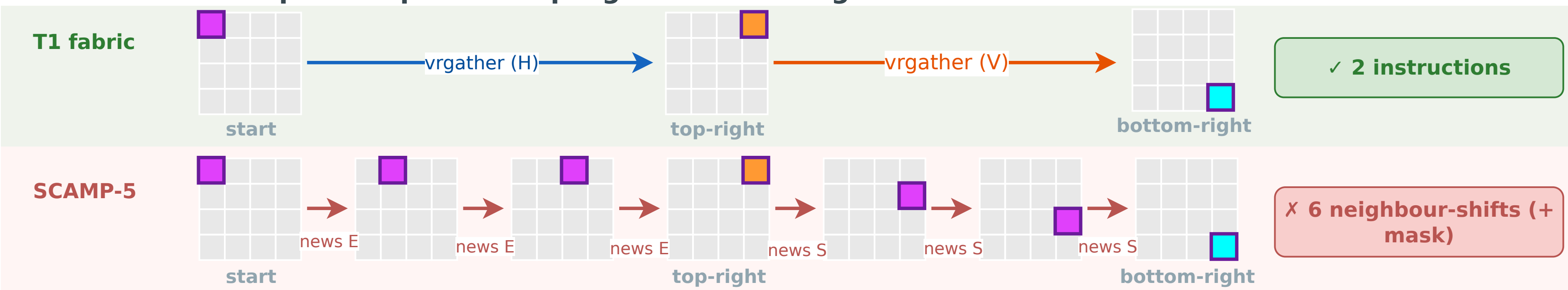
Programmability: T1 2D fabric vs SCAMP-5 pixel-processor array

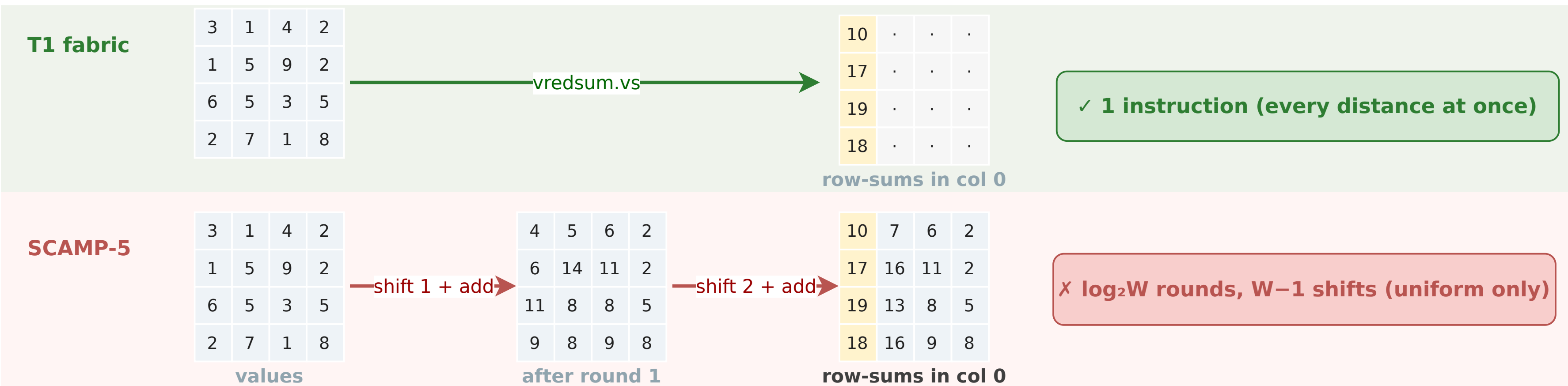
SCAMP-5 has one analog processor per pixel but its ONLY links are the 4 nearest neighbours (N/E/W/S), run in lock-step (SIMD). Non-local moves must walk cell-by-cell; Non-uniform ops (each pixel a different distance from the target) need many shift rounds. T1 reaches any lane directly (gather / slide / H-V transpose) and reduces in hardware.



Task A · move one pixel: top-left → top-right → bottom-right



Task B · sum each row into the left column (vredsum) — an non-uniform move: every pixel is a different distance from the target



Summary

Aspect	SCAMP-5	T1 2D fabric
Inter-PE connectivity	4 nearest neighbours (N/E/W/S)	any pixel in same row/col → any pixel in same row/col (gather) + H/V transpose
Relocate a pixel by (dx,dy)	|dx|+|dy| sequential shifts	1 gather (vrgather)
Sum each row	$\log_2 W$ shift-add rounds, all PEs shift together	1 vredsum (hardware reduction)
Control granularity	SIMD: every pixel moves the SAME way without masking	High level lane process control, uniform & non-uniform, regular & irregular ops